

Java OOP Lecture Notes

JAVA OBJECT ORIENTED PROGRAMMING LECTURE NOTES

Chapter 1: Classes and Objects

A class is a user-defined blueprint or prototype from which objects are created. It represents

An object is a basic unit of Object-Oriented Programming and represents real-life entities. An

In Java, memory for objects is dynamically allocated in the Heap using the 'new' keyword.

Chapter 2: Constructors

A constructor in Java is a special member function used to initialize objects. It is called whe

Key rules for constructors:

1. Constructor name must match the class name exactly.
2. A constructor must have NO explicit return type (not even void).
3. If no constructor is written, Java provides a default no-argument constructor.

Chapter 3: Inheritance

Inheritance is the mechanism in Java by which one class acquires the properties (fields) and be

The 'extends' keyword is used to inherit from a class. The class that inherits is called the su

Java supports single inheritance and multilevel inheritance for classes, but NOT multiple inher

Chapter 4: Method Overriding

If a subclass provides the specific implementation of a method that has already been declared i

Rules for Method Overriding:

1. The method in the child class must have the exact same name as in the parent class.
2. The method in the child class must have the exact same parameter list and compatible return
3. The access modifier cannot be more restrictive than the parent class method (e.g. if supercl
4. Static methods, private methods, and final methods CANNOT be overridden.

Method overriding is used for Runtime Polymorphism (dynamic method dispatch).

Chapter 5: Abstract Classes

An abstract class in Java is a class declared with the 'abstract' keyword. It cannot be instant

It may contain abstract methods (methods declared without an implementation body) as well as co

If a regular class extends an abstract class, it must implement all abstract methods, otherwise

Chapter 6: Interfaces

An interface in Java is a blueprint of a class. It is declared using the 'interface' keyword an

Interfaces specify WHAT a class must do, but not HOW it does it.

All fields in an interface are implicitly public, static, and final.

All methods in an interface are implicitly public and abstract (prior to default/static methods

A class can implement multiple interfaces, allowing Java to achieve multiple inheritance of typ